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Anticorruption, Government Subsidies, and Innovation: Evidence from China

Lily Fang,^a Josh Lerner,^{b,*} Chaopeng Wu,^c Qi Zhang^d
^a Finance Department, Institut Européen d'Administration des Affaires (INSEAD), Fontainebleau, France 77305; ^b Entrepreneurial Management Unit and Finance Unit, Harvard Business School, Boston, Massachusetts 02163; ^c School of Management, Xiamen University, Xiamen, Fujian 361005, China; ^d School of Management, Shandong University, Jinan, Shandong 250100, China

*Corresponding author

Contact: lily.fang@insead.edu,  <https://orcid.org/0000-0003-1673-7368> (LF); jlerner@hbs.edu,  <https://orcid.org/0000-0003-3822-2162> (JL); wuchaopeng@xmu.edu.cn,  <https://orcid.org/0000-0002-2298-5950> (CW); qizhang@sdu.edu.cn,  <https://orcid.org/0000-0002-0673-0421> (QZ)

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Abstract. We leverage an exogenous shock—the crackdown on corrupt Chinese officials beginning in 2012—and examine how the allocation of research subsidies and innovative outcomes were affected. We argue that the staggered removal of provincial heads on corruption charges during China's anticorruption campaign and the unanticipated departures of local government officials responsible for innovation programs led to plausibly exogenous reductions in corruption. After both events, the allocation of subsidies became more sensitive to firm merit than to corruption and subsidies became more strongly associated with future innovation. Anticorruption efforts and officials' career incentives improved the efficacy of subsidy programs.

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Keywords: research and development • innovation • government • subsidies • anti-corruption

1. Introduction

Innovation and research and development (R&D) are vital to economic growth. However, because of the positive externalities generated by R&D, the private sector may underinvest (Nelson 1959, Arrow 1962). As a result, governments frequently subsidize R&D activities: Among major economies, countries spend between 0.02% of gross domestic product (GDP; Japan) to 0.38% of GDP (Russia) on R&D subsidies (OECD 2021). In recent years, economists have become increasingly interested in understanding the impact of public subsidies on innovation, but the evidence appears mixed. On the one hand, using U.S. data, Howell (2017) finds that research grants spur firm growth by funding technology prototyping. On the other hand, using Chinese data, Wang et al. (2017) find that government funding has no causal effect on innovation.

In this paper, we examine how corruption affects the relationship between subsidies and innovation. We do this in the context of China, a country that—as we discuss in Section 2.1—both heavily subsidized its private sector R&D and had widespread corruption. At the same time, the promotion of government officials depended heavily on their ability to stimulate local economic growth.

We argue that in China corruption and career concerns acted as two opposing influences on local officials' subsidy-granting decisions. We hypothesize that corruption impeded the effectiveness of R&D subsidy programs. When corruption was unchecked, fund allocation tended to reflect corruption rather than merit, and the subsidies had a weak relationship with future innovation. Allocation effectiveness improved when corruption was reduced.

To empirically test these hypotheses, we exploit the sweeping anticorruption campaign that Chinese president Xi Jinping unleashed in 2012 and the unanticipated departures of local officials as two identification channels that reduced corruption in a plausibly exogenous fashion. (Section 2.1 provides details on these approaches). Using difference-in-differences methods around these two types of events, we examine changes in the way R&D subsidies were allocated and changes in the relationship between R&D subsidies and future innovation.

We find that prior to either the anticorruption campaign or unanticipated rotations of innovation officials, corruption (proxied by a firm's abnormal entertainment and travel costs) and merit (proxied by its R&D efficiency) both affected the number of subsidies that firms

received. On average, the two inputs had roughly equal influence: A one-standard-deviation increase in either variable led to a roughly 10% increase in subsidies as a percentage of revenue received. However, in a difference-in-differences analysis, we find that both the anticorruption campaign and the rotation of provincial innovation officials affected the influence of these two factors, but in opposite ways. These events, which we argue were plausibly exogenous shifts that reduced the prevalence of petty corruption, sharply increased the relative impact of merit and simultaneously decreased the influence of corruption on subsidies. These effects were particularly dramatic for firms with high R&D efficiency and those with high ratios of abnormal entertainment and travel costs to sales ratio.

Postevent years also saw a stronger relationship between R&D subsidies and future innovation, an effect that does not seem to be dependent on a particular specification.

In cross-sectional analysis, we find that the key patterns—the increasing relationship between merit and subsidies and the decreasing relationship with abnormal entertainment expenses and subsidies—were particularly strong in (a) industries with a high dependence on external financing and (b) in areas where there were fewer banks and venture capital firms. The effects were also particularly strong for subsidies that strongly related to innovation. We also find evidence suggesting a “deterrence effect”: After the inception of the anticorruption campaigns in 2012, the sensitivity of subsidies to corruption fell across the board, including provinces where government officials were not replaced. The effect was also seen among new officials who were promoted within the same province, suggesting the effect was not a consequence of severed social networks.

We explore a wide variety of robustness checks. These include, among others, (a) creating alternative measures of R&D efficiency, using different patent-based metrics based on Chinese, U.S., and World Intellectual Property Office filings, and (b) using propensity score matching to control for differences between the firms that did and did not receive subsidies.

In conclusion, we should note that we do not argue that government subsidies “caused” firm-level innovation. Our focus is instead to document how corruption weakened the link between subsidies and innovation. In doing so, our paper speaks broadly to the impact of corruption on development. One strand of political science literature argues that corruption is largely harmless and acts to “grease the wheel” of commerce (Huntington 1968). Although the setting around Chinese anticorruption campaign had its unique aspects, our findings are more in line with those of Bertrand et al. (2007), who argue that corruption creates real economic distortions. These effects seem particularly striking among high-performing research organizations: Given the well-documented skewness of innovative outcomes, this finding underscores the potential importance of these distortions.

2. Empirical Design

2.1. Backdrop

President Xi’s anticorruption campaign is an attractive empirical setting to study the effect of corruption on R&D subsidies and innovation. As we describe later, the campaign represented a sudden shock and led to the removal of many provincial leadership figures. At the same time, the Chinese setting during the period had unique institutional features, which, although interesting, may limit the generalizability of the results.

First, the allocation of research subsidies in China was different from that in many Western nations (for more background, see Ding and Li (2017)). The funds allocated were substantial: One fifth of Chinese firms’ R&D expenditures came from government subsidies. However, unlike in the West, where research grants have typically been evaluated by independent scientific panels for many decades, the allocation of R&D subsidies in China was controlled by layers of government officials.

In particular, since the 1990s, China has had technology bureaus at each level of government—central, provincial, and municipal—responsible for promoting innovation. The labyrinth of technology bureaus offered a variety of subsidies out of their fiscal budgets, including funds for major R&D projects, commercialization, projects by small and medium enterprises specifically, and even patent application fees. Firms applied for subsidies at the lowest (municipal) level. Municipal technology bureaus could award funding from their R&D subsidy budget and/or pass the application to the provincial level bureau for significantly larger amounts of funding. Provincial-level bureaus in turn made awards from their own budgets and occasionally passed the applications up to the central government level. Applications that reached the central government level were subject to heavy scrutiny. In contrast, at the municipal or provincial levels, the bureau officials had almost complete discretion over funding decisions.

This setup put local government officials—who were almost never scientific experts—in a convenient position for rent-seeking behavior. It also created a strong incentive for Chinese firms to cultivate good “guan xi” (relationships) with these local officials through bribery. The amount of money at stake was large; according to the 2021 *China Statistical Yearbook* (National Bureau of Statistics of China 2021), in 2020, subnational-level technology bureaus were collectively in charge of 580 billion yuan (91.4 billion USD) in annual funding, representing 64.3% of the total government R&D subsidies; of this sum, provincial-level bureaus controlled the lions share. In China’s nine-level hierarchy of government officials, provincial technology bureau heads sat at “Level Five,” exactly in the middle. It was well documented in the Chinese literature that midlevel petty corruption is the most widespread form of corruption in China.¹ For these reasons, our empirical analysis focuses on corruption and R&D subsidies at the provincial level.

Two other Chinese institutional features motivate our identification strategies. Our first identification approach is based on the anticorruption campaign waged by President Xi Jinping starting in 2012. The sweeping and sudden nature of the campaign was a shock to the system; the onset of the campaign was neither controlled by individual firms nor anticipated. Figure 1 traces the frequency of Chinese newspaper articles that contain the phrase “anti-corruption” in the title.² It shows a sharp increase in media mentions of anticorruption in 2012, the first year of the anticorruption campaign. The central government’s campaign against corruption was staggered over time across different provinces, allowing for a richer identification.

Between 2012 and 2018, Xi’s administration charged 42 top provincial officials (provincial governors and vice governors and provincial party secretaries and vice secretaries) across 23 provinces and removed them from their posts. These high-profile cases were publicized on the Central Commission for Discipline Inspection (CCDI) website.³ Table 1 summarizes the year, province, and number of officials removed based on data from the CCDI website. Throughout this paper, unless otherwise noted, we refer to these events as the removals of top provincial officials on corruption charges.

Although some of these corruption charges might have been politically motivated, a growing body of empirical evidence indicates that the anticorruption campaign had a deterrent effect on China’s business culture (Griffin et al. 2022, Li et al. 2018, Chen and Kung 2019, among others). The sharp drop since 2012 in entertainment and travel costs (ETC) spending, which as discussed later, has been often associated with corruption, evidenced in Figure 2, corroborates this point.

As a second identification approach, we used the unanticipated departures of provincial-level technology

bureau heads. According to the “Party and Government Leading Cadres Tenure Regulations” codified in 2006,⁴ technology bureau heads were required to step down after a five-year term. However, sometimes departures occurred outside this schedule, due to retirements, health issues, investigations, or the Party’s needs to rotate its cadres. To capture these “out-of-cycle” and thus unanticipated departures, we focus on departures that took place at least one year before the anticipated succession date. These sudden departures broke the cozy relationships between the firms and the officials, leading to a plausibly exogenous reduction in corruption after departure.

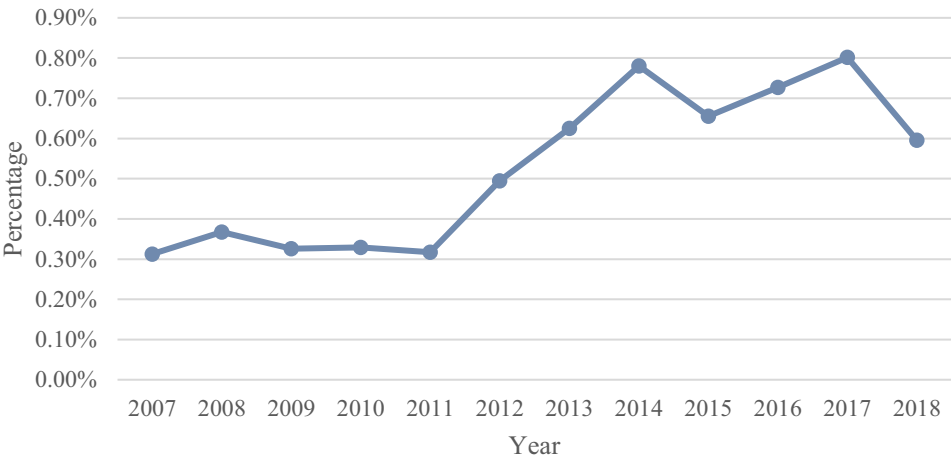
Figure 3 shows the number of unanticipated departures of provincial technology bureau heads from 2007 to 2018. The departures occurred both before and after the anticorruption campaign in 2012; they thus provide a separate identification for our analyses. We manually compile the list of departures from the official websites of each provincial government and the central government, newspaper reports, and postings and announcements about personnel movements from the Chinese Communist Party’s Organization Department.⁵ In total, 52 cases of unanticipated official departures in 24 provinces in these years fit our criteria.

At the same time, an extensive amount of political science literature (Li and Zhou 2005) suggests Chinese officials’ promotion prospects depended on their track records, far more so than in other countries. Career concerns thus created incentives for the government officials responsible for innovation programs to reward the most promising firms and allocate subsidy funds according to firm merits.

2.2. Measurements of Key Variables

Our central hypothesis is that government officials were motivated by private benefits through corruption on the

Figure 1. (Color online) Media Mentions of Anticorruption

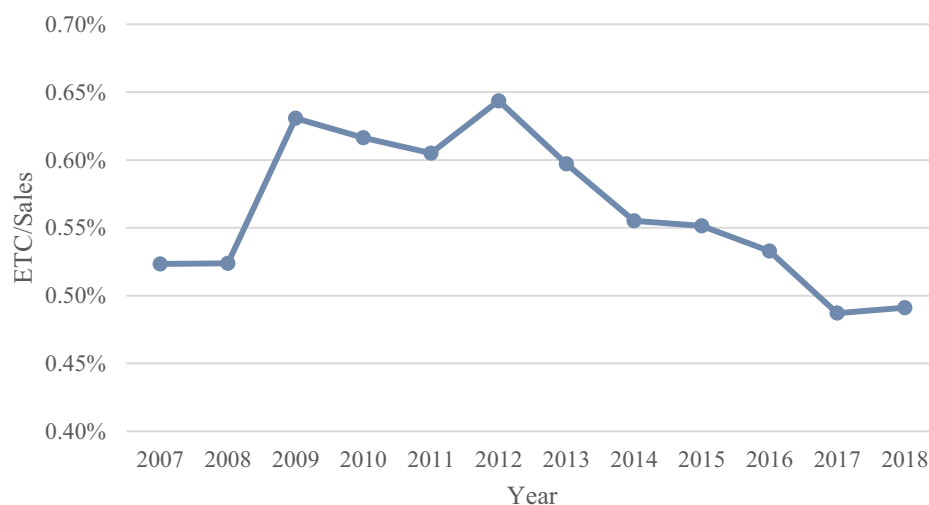


Note. This graph shows the percentage of articles in all official provincial newspapers (the “Daily” newspapers published by provincial governments) with the words “anti-corruption” in their titles over time.

Table 1. Removal of Top Provincial Officials on Corruption Charges During the Anticorruption Campaign

Year	Province	Names of officials removed	Number of officials removed
2012	Sichuan	Li, Chuncheng	1
2012	Chongqing	Wang, Lijun; Bo, Xilai	2
2013	Anhui	Ni, Fake	1
2013	Hubei	Guo, Youming	1
2013	Sichuan	Guo, Yongxiang	1
2014	Hainan	Ji, Wenlin; Tanli	2
2014	Yunnan	Shen, Peiping	1
2014	Jiangxi	Yao, Mugen	1
2014	Shanxi(山西)	Du, Shanxue; Ren, Runhou	2
2014	Inner Mongolia	Pan, Yiyang	1
2015	Fujian	Xu, Gang; Su, Shulin	2
2015	Hebei	Zhou, Benshun	1
2015	Jilin	Gu, Chunli	1
2015	Shanghai	Ai, Baojun	1
2015	Ningxia	Bai, Xueshan	1
2015	Beijing	Lyu, Xiwen	1
2015	Yunnan	Qiu, He	1
2016	Anhui	Yang, Zhenchao; Chen Shulong	2
2016	Guangdong	Liu, Zhigeng	1
2016	Jiangsu	Li, Yunfeng	1
2016	Sichuan	Li, Chengyun	1
2016	Tianjin	Yin, Hailin; Huang, Xingguo	2
2017	Gansu	Yu, Haiyan; Wang, Sanyun	2
2017	Anhui	Zhou, Chunyu	1
2017	Chongqing	Sun, Zhengcai; Mu, Huaping; He, Ting	3
2017	Liaoning	Liu, Qiang	1
2018	Shannxi(陕西)	Feng, Xinzhu	1
2018	Shandong	Ji, Xiangqi	1
2018	Jiangxi	Li, Yihuang	1
2018	Guizhou	Wang, Xiaoguang; Pu, Bo	2
2018	Inner Mongolia	Bai, Xiangqun	1
2018	Jiangsu	Miao, Ruilin	1
Total	23		42

Notes. This table summarizes the removal of top provincial officials on corruption charges after the inception of the anticorruption campaign. Included are removals of provincial governors or vice governors and Party secretaries or vice secretaries. The data are compiled from the website of the Central Commission for Discipline Inspection.

Figure 2. (Color online) Average ETC Spending over Time

Notes. This figure shows the trend of the average ETC as a percentage of sales. ETC is firms' entertainment and travel costs, as reported in their annual statements.

Figure 3. (Color online) Unanticipated Departures of Provincial Technology Bureau Heads



Note. This figure shows the number of unanticipated departures of provincial technology bureau heads by year.

one hand and career concerns on the other. We have two empirical predictions: first, subsidy grant decisions were affected by both corruption and by firm merit; second, when corruption was rampant, subsidies had a weaker relationship with future innovation than when corruption was more in check. This section discusses our sample and construction of the key variables.

Our sample consists of Chinese firms listed in the Shanghai and Shenzhen stock exchanges. We focus on firms in the following sectors (the Chinese Securities Regulation Commission (CSRC) industry codes in parentheses), which Jaffe and Trajtenberg (2002) argue are relatively R&D intensive: petrochemicals (C4), electronics (C5), metals and materials (C6), machinery and equipment (C7), pharmaceuticals and biotechnology (C8), and information technology (G). Firms in these sectors represent 56% of the total market capitalization of China's A-share market (excluding the financial sector) and 82% and 89% of their total R&D expenses and U.S. patents, respectively, during our sample period of 2007 through 2018.

To proxy for firm-level corruption, we use the ETC reported by Chinese firms in the footnotes of their annual statements. Cai et al. (2011) document that, although ETC contained legitimate business expenses, there was significant latitude in how employees claimed expenses, making this accounting item indicative of corruption. For example, gifts, alcohol, and banquets used as bribes were frequently procured at business hotels and expensed as ETC. (Internet Appendix 1 contains several validation tests on ETC as a corruption measure.) The type of corruption that ETC represented was "petty corruption" by midlevel officials, which is precisely what we want to capture. It was the type of corruption that represented "side payments," which may have "greased the wheel" of commerce. There may well have been more significant forms of corruption elsewhere, but petty corruption nevertheless could influence midlevel officials with significant decision power over the allocation of resources.

Figure 2 plots the average ETC spending among our sample firms over time. We observe a sharp drop in ETC after 2012, the onset of the anticorruption campaign. On average, firms spent 0.6% of total sales on ETC, which was roughly one seventh of the firms' R&D spending (4.1% of sales). Thus, in the aggregate, petty corruption added up and consumed a nontrivial portion of GDP.

One problem with the raw ETC data is that they are a noisy measure of petty corruption. To control for factors that affect legitimate business costs, we follow Cai et al. (2011) and the accounting literature on the treatment of discretionary accruals (Kothari et al. 2005, Gul et al. 2011) and estimate the following cross-sectional regression for each industry-year subsample:

$$ETC/Sales_{i,t} = \gamma_0 + \gamma_1 \ln(Assets)_{i,t} + \gamma_2 Business\ In\ Other\ Regions_{i,t} + \gamma_3 PerCapGDP_{i,t} + \varepsilon_{i,t}, \quad (1)$$

where $\ln(Assets)$ is the log of book value of a firm's total assets, *Business In Other Regions* is the number of geographical regions where a firm's revenue comes from other than the region where the firm is based,⁶ and *PerCapGDP* is the log of per capita GDP of the firm's home province. We use these three control variables to estimate firms' predicted $ETC/Sales$, which is likely to vary systematically according to legitimate business needs.⁷ We then take the residual from this regression as the abnormal $ETC/Sales$ ($AETC/Sales$) incurred by the firm, which we use as the primary proxy for firm-level corruption in the analyses.

For firm-level R&D subsidies, we manually collect data from the footnotes of companies' annual reports. Rule 16 of the Accounting Rules of China's Enterprises announced in 2006 required listed firms to disclose the type and amount of government subsidies received. For each firm-year in our sample, we calculate the total R&D subsidies (scaled by revenue) as the sum of the following seven types of funding disclosed in filings:

Table 2. Categories of R&D Subsidies

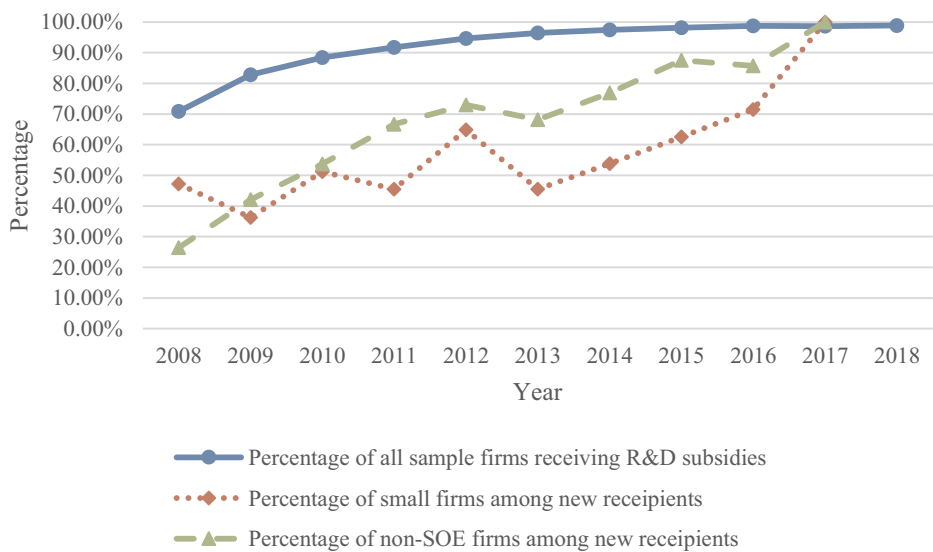
Funding category	Subsidy description	Total amount (RMB, mil)	Percentage of total
1	Subsidies for product development, intermediate testing, and major R&D projects	2,445	2.07%
2	National and provincial Small and Medium Technology Enterprises Innovation Funds (InnoFund)	141	0.12%
3	Subsidies for small and medium enterprises' technological adaptation and upgrading	127	0.11%
4	Subsidies for technological modification and upgrading	11,206	9.47%
5	Subsidies for technology commercialization and equipment and systems purchase	31,089	26.29%
6	R&D grants	72,401	61.22%
7	Subsidies for patent applications	858	0.72%
	Total	118,269	100%

Notes. This table tabulates the seven categories of R&D subsidies collected from the footnotes of companies' financial reports. We report the aggregate amount and percentage share of the subsidies.

- (1) Subsidies for product development, intermediate testing, and major R&D projects;
- (2) Funding from the national and provincial Small and Medium Technology Enterprises Innovation Funds (also known as InnoFunds);
- (3) Subsidies for small and medium enterprises' technological adaptation and upgrading;
- (4) Subsidies for technological modification and upgrading;
- (5) Subsidies for technology commercialization and equipment and systems purchase;
- (6) R&D grants; and
- (7) Subsidies for patent applications.⁸

Table 2 lists the breakdown of these seven categories. The largest sources of subsidies were R&D grants (category 6) and subsidies for commercialization (category 5). The InnoFund (category 2), which was earmarked for small unlisted firms and has attracted recent research interest (Wang et al. 2017), represented a small part of the total subsidies for our sample of large listed firms. Subsidies were pervasive in China. Figure 4 shows that by 2018, 99% of the firms in our sample had received R&D subsidies. The figure also shows that over the years, an increasing fraction of small firms (defined as those with below-median market cap in each year) and non-state-owned enterprises (SOEs) received subsidies.

Figure 4. (Color online) Subsidies Among Sample Firms



Notes. This figure plots the percentages of different types of firms receiving R&D subsidies over time. The solid line shows the percentage of firms in our sample that received subsidies as of a given year. The dashed line shows the percentage of small firms (firms whose total assets were less than the median value in the sample in a given year) among the new recipients. The dotted line shows the percentage of non-SOE firms among new recipients. There were no new recipients in 2018 in our sample. The SOE classification is detailed in Appendix B.

Table 3. Sample Descriptive Statistics

Variable	Mean	Standard deviation	Minimum	Median	Maximum	Observations
Panel A: Main variables						
<i>Subsidies/Sales</i>	0.005	0.008	0.000	0.002	0.057	12,158
<i>ETC/Sales</i>	0.006	0.008	0.0001	0.004	0.075	12,158
<i>AETC/Sales</i>	0.000	0.007	−0.017	−0.001	0.070	12,158
<i>R&D/Sales</i>	0.041	0.047	0.000	0.033	0.349	12,158
<i>Subsidies/R&D</i>	0.191	0.488	0.000	0.070	4.914	10,761
<i>Patents/Sales (U.S.)</i>	0.0001	0.001	0.000	0.000	0.006	12,158
<i>Relative Citation Strength (U.S.)</i>	0.037	0.213	0.000	0.000	1.855	12,158
<i>R&D Efficiency</i>	0.046	0.117	0.000	0.004	1.058	11,905
Panel B: Control variables						
<i>Size (Mil. RMB)</i>	5781	10,978	166.97	2,452	92,309	12,158
<i>Age (Year)</i>	16.84	5.265	2	16	41	12,158
<i>Leverage</i>	0.414	0.224	0.033	0.401	1.412	12,158
<i>Return on Assets</i>	0.034	0.074	−0.470	0.037	0.247	12,158
<i>Tobin's q</i>	2.509	2.150	0.233	1.912	16.83	12,158
<i>Intangible Assets/Assets</i>	0.045	0.040	0.000	0.035	0.283	12,158
<i>SOE</i>	0.344	0.475	0	0	1	12,158
<i>Political Connection</i>	0.221	0.415	0	0	1	12,158
<i>Business in Other Regions</i>	5.309	2.519	0	5	8	12,158

Notes. This table provides descriptive statistics of our sample, which consists of annual firm-level observations between 2007 and 2018. Detailed variable definitions are found in Appendix B. All financial amounts are measured in millions of RMB. All financial ratios are calculated annually using data from firms' annual statements.

To measure a firm's "merit" for an innovation subsidy, we use a measure that reflects its R&D efficiency. Similar to Hirshleifer et al. (2013), we define firms' R&D efficiency as

$$R\&D\ Efficiency_{i,t} = Patent_{i,t} / (R\&D_{i,t} + 0.8 \times R\&D_{i,t-1} + 0.6 \times R\&D_{i,t-2}), \tag{2}$$

where $Patent_{i,t}$ in our baseline analysis reported in this paper is firm i 's number of new invention patent⁹ applications filed with the Chinese State Intellectual Property Office in year t and approved by August 31, 2021; and the $R\&D_{i,t}$, $R\&D_{i,t-1}$, and $R\&D_{i,t-2}$ are the R&D expenditures in millions of RMB during year t , $t - 1$, and $t - 2$ from firms' annual statements.¹⁰ The data on Chinese invention patents came from the InnoJoy patent database, a commercial database providing patent information for Chinese firms, including a patent's type, application year, grant year (if granted), assignee name, and the patent's technological class and citation information. (For more detail on patent data and variable construction, see Appendix A.) Statistics in Table 3 show that the average *R&D Efficiency* measure was 0.046, meaning that on average, it took about 21.7 (1/0.046) million RMB (~3.4 million USD) of R&D capital to generate one Chinese invention patent.

For firms' innovative output, we use two measures based on firms' U.S. utility patents¹¹: *Patents/Sales* and *Relative Citation Strength*. We use U.S. utility patent measures for future innovation because we want these measures to capture significant innovations. We obtain U.S. utility patent information by searching the names of all Chinese listed firms and their wholly owned subsidiaries

from the InnoJoy patent database. *Patents/Sales* is the number of U.S. utility patent applications filed by a firm in year t that are ultimately granted by August 31, 2021, divided by its revenue in year t . For each patent, we calculate its *Relative Citation Strength* as the number of citations it received up to August 31, 2021, divided by the average number of citations per patent received over the same period by all patents applied for in the same application year t as the patent in question (that are also ultimately granted by August 31, 2021) and in the same primary four-digit technological class. This scaling controls for the application year and technological class. A *Relative Citation Strength* higher than one means that a patent is cited more than the average patent successfully filed for in the same year in the same technology class.

Two issues with this measure should be acknowledged. First, many firms in our sample do not file U.S. patents. In the 2007–2018 period that we analyze, 18.1% of the firms patent in the United States at any point and 87% do so in China at any point.¹² As a robustness check, we repeat the key analyses below using counts of Chinese patents.

A second concern with all patent-related measures is truncation. Some patent applications filed during the latter years in our sample had not been granted by August 31, 2021. We correct for this truncation bias by multiplying the number of patent applications filed by a firm in year t by a weighting index using the methodology of Hall et al. (2005) and Lerner and Seru (2022). Also, we include year fixed effects in our regressions throughout to further alleviate this problem. Our results are qualitatively unchanged if we used unadjusted patent measures.

Table 4. Pearson Correlation Matrix

	1	2	3	4	5	6	7	8	9	10	11	12	13
1. Subsidies/Sales	1												
2. R&D Efficiency	0.045***	1											
3. AETC/Sales	0.071***	-0.024***	1										
4. Patents/Sales (U.S.)	0.146***	0.081***	-0.011	1									
5. Relative Citation Strength (U.S.)	0.084***	0.046***	-0.004	0.597***	1								
6. Size	-0.092***	-0.040***	0.057***	-0.012	0.086***	1							
7. Age	-0.029***	-0.113***	0.025***	-0.021**	0.006	0.144***	1						
8. Leverage	-0.215***	-0.026***	0.055***	-0.078***	-0.021**	0.290***	0.164***	1					
9. Return on Asset	0.027***	0.035***	-0.118***	0.040***	0.049***	-0.027***	-0.125***	-0.397***	1				
10. Tobin's q	0.126***	0.003	0.129***	0.020**	-0.014	-0.242***	0.008	-0.225***	0.124***	1			
11. Intangible Assets/Assets	0.024***	0.026***	0.084***	-0.009	0.001	-0.052***	0.049***	0.095***	-0.105***	0.060***	1		
12. SOE	-0.091***	0.042***	-0.049***	-0.053***	-0.031***	0.233***	0.084***	0.311***	-0.107***	-0.185***	-0.033***	1	
13. Political Connection	0.043***	0.030***	-0.015*	0.028***	0.018**	-0.011	-0.041***	-0.083***	0.051***	0.013	0.036***	-0.178***	1

Note. This table presents the Pearson correlation matrix among the main variables in our analysis.
*, **, ***Statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 3 provides summary statistics for our sample, which consists of annual firm-level observations between 2007 and 2018. Several observations can be made. First, on average firms spent roughly 0.6% of total sales on ETC. Because the average net profit margin in our sample was 5.3%, ETC spending was significant. From unreported statistics, we know that firms on average received 0.5% of annual sales as R&D subsidies. If we assume that 10% of the ETC spending was corruption, then the ratio between subsidies received to money spent on corruption would be about 8.5 ($5/(0.1 \times 6)$). This suggests that petty corruption was a fairly efficient way for firms to obtain resources.¹³ Second, subsidies were an important source of overall R&D funding, representing 19.1% of our sample firms' total R&D spending. This magnitude matches almost exactly the 20.2% reported in the 2018 China Statistical Yearbook (National Bureau of Statistics of China 2018) and suggests that our sample was representative of Chinese firms. Third, Chinese firms made substantial R&D investments: on average, the annual R&D expenditures were 4.1% of sales in our sample, higher than the 3.5% R&D/Sales figure reported for U.S. firms (including basic industries not in the Chinese sample).¹⁴ The unconditional Relative Citation Strength for the firms' U.S. patents was only 0.037, which is due to the relative rarity of U.S. patents among Chinese firms. Among granted U.S. patents, the Relative Citation Strength (unreported) was 0.74.

Table 4 presents the results of Pearson correlation tests among the main variables in our sample. Firm-level R&D efficiency and corruption (AETC) were both highly correlated with R&D subsidies, and R&D subsidies was highly correlated with firms' future innovation. Corruption and R&D efficiency were negatively correlated.

Finally, to identify the effect of corruption on the subsidy-innovation relationship, we rely on two types of events that changed the amount of corruption in a plausibly exogenous way. As discussed in Section 2.1, these are the removal of a leading official in a given province on corruption charges as part of central government's campaign against corruption and the unanticipated departures of provincial-level technology bureau heads.

One concern relating to both types of events is that they may be correlated with the local economic conditions and firm performance. We conducted probit analysis of these events using lagged provincial GDP growth, local firms' average return on assets, and patent counts (scaled by sales) as proxies of local economic conditions. None of these variables had explanatory power for either the removals or the departures. These results are reported in Table IA2 in the online appendix.

3. Subsidy-Granting Decisions: Firm Merit vs. Corruption

In this section, we empirically examine our hypothesis that both firm-level corruption and firm merit affected

the amount of subsidies a firm received but that the weight of the two variables differed depending on the extent of corruption.

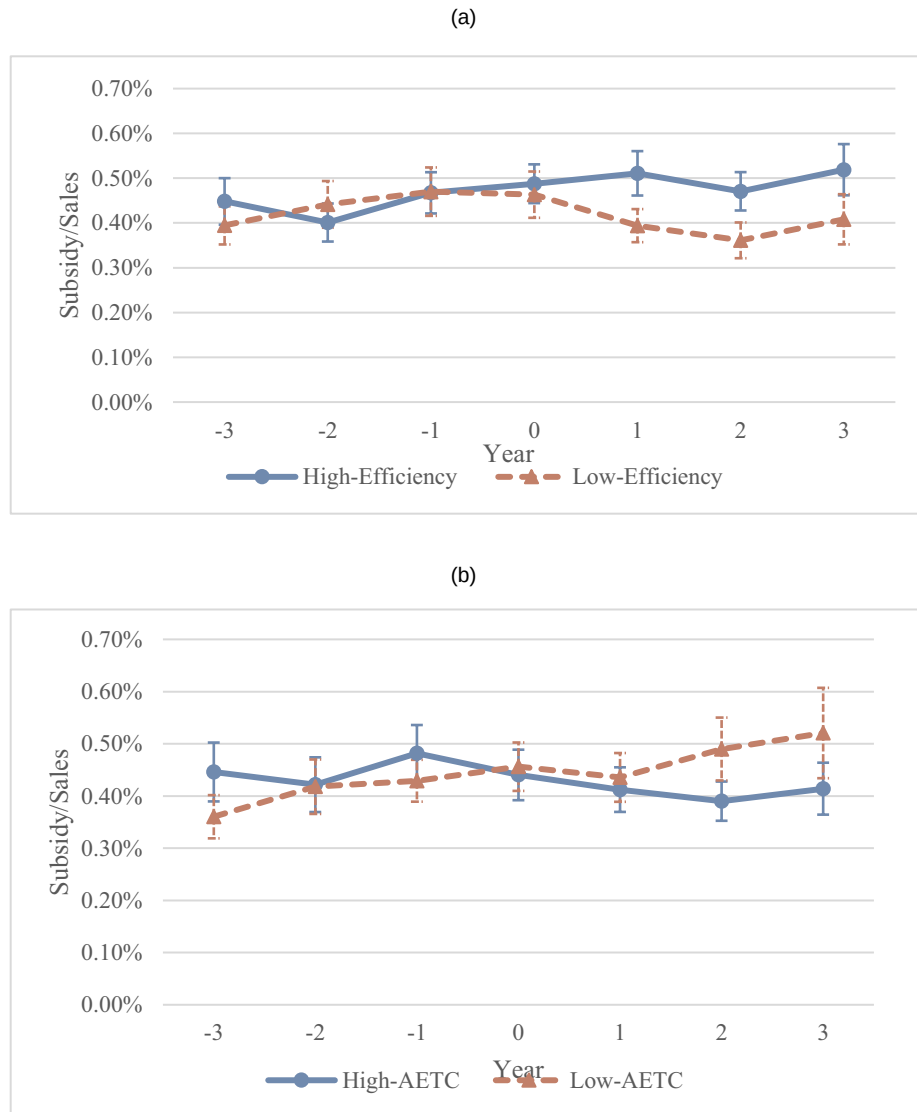
Figures 5 and 6 graphically illustrate our baseline difference-in-difference results.¹⁵ Figures 5(a) and 6(a) show that, postevent, subsidies to low-efficiency firms were virtually unchanged, but subsidies to high-efficiency firms increased substantially. Figures 5(b) and 6(b) show, postevent, subsidies to high-AETC firms were virtually unchanged, but subsidies to low-AETC firms increased substantially. Together these patterns indicate that corruption had less impact on subsidies

postevent, whereas firm merit had more impact, consistent with our hypothesis.

In the multivariate setting, we estimate the following panel regression:

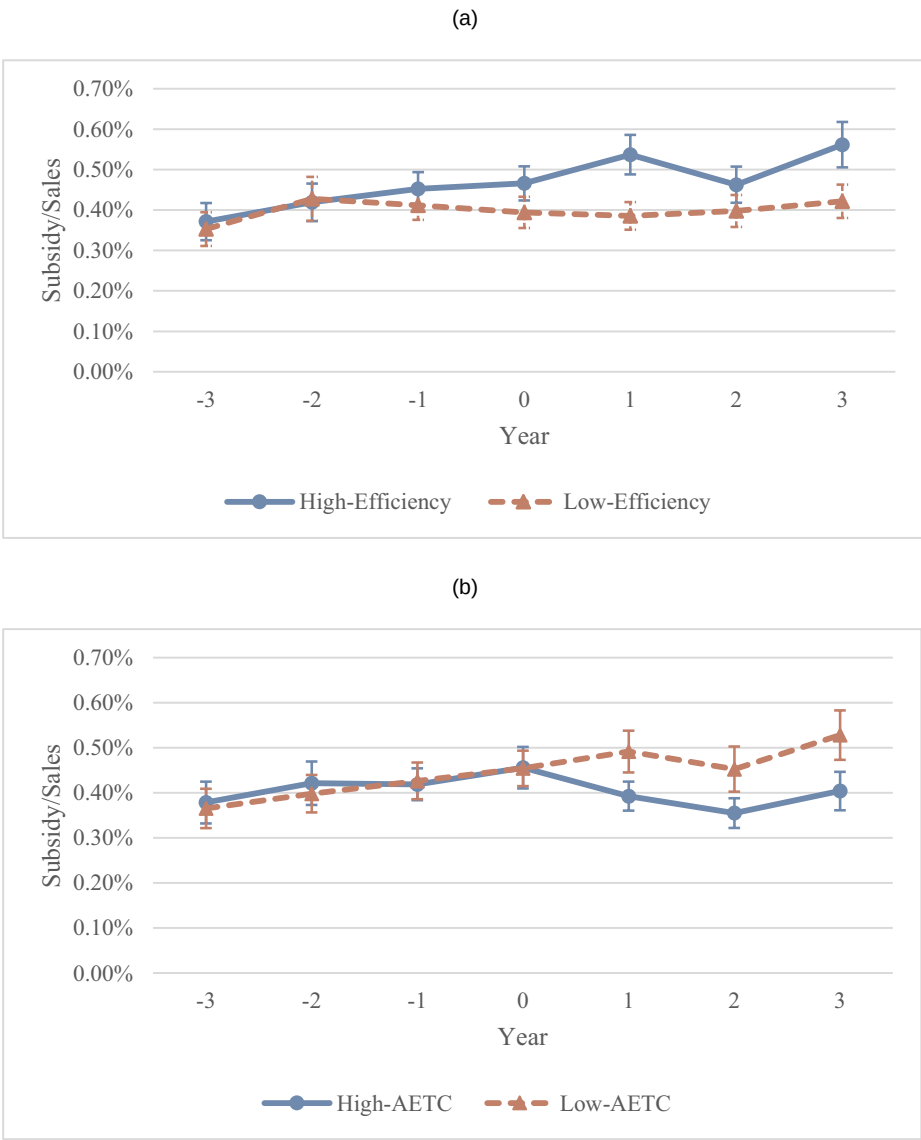
$$\begin{aligned} \text{Subsidies}/\text{Sales}_{i,t+1} = & \beta_0 R\&D\text{ Efficiency}_{i,t} + \beta_1 AETC/\text{Sales}_{i,t} \\ & + \beta_2 R\&D\text{ Efficiency}_{i,t} \times \text{Post Event}_{i,t} \\ & + \beta_3 AETC/\text{Sales}_{i,t} \times \text{Post Event}_{i,t} \\ & + \beta_4 \text{Post Event}_{i,t} + \text{Controls} \\ & + \text{Fixed Effects}. \end{aligned} \quad (3)$$

Figure 5. (Color online) Subsidies Before and After Removal of Top Provincial Officials on Corruption Charges



Notes. This figure plots the R&D subsidies (scaled by firm revenue) received by firms before and after the removal of top provincial officials on corruption charges. (a) Firms are sorted by the average *R&D Efficiency* (calculated according to Equation (2)) during the three event years prior to the removals of top provincial officials. Firms with above- (below-) median *R&D Efficiency* in a given year are classified as high- (low-) efficiency firms. We match each high-efficiency firm with one low-efficiency firm within the same market capitalization quartile and book to market ratio quartile in the year prior to the events and having the smallest difference in subsidies over the three-year period prior to the events. (b) Firms are sorted by the average *AETC/Sales* during the three event years before the removal of top provincial officials. Firms with above- (below-) median spending are classified as high- (low-) AETC firms. We use the same matching procedure as described above. The vertical bars represent standard errors.

Figure 6. (Color online) Subsidies Before and After Unanticipated Departure of Provincial Technology Bureau Heads



Notes. This figure plots the R&D subsidies (scaled by firm revenue) received by firms before and after the unanticipated departure of provincial technology bureau heads. (a) Firms are sorted by the average *R&D Efficiency* (calculated according to Equation (2)) during the three event years prior to the departures of provincial technology bureau heads. Firms with above- (below-) median *R&D Efficiency* in a given year are classified as high- (low-) efficiency firms. We match each high-efficiency firm with one low-efficiency firm within the same market capitalization quartile and book to market ratio quartile in the year prior to the events and having the smallest difference in subsidies over the three-year period prior to the events. (b) Firms are sorted by the average *AETC/Sales* during the three event years before the removal of top provincial officials. Firms with above- (below-) median spending are classified as high- (low-) AETC firms. We use the same matching procedure as described previously. The vertical bars represent standard errors.

The dependent variable is the amount of R&D subsidies received by a firm in a given year (scaled by sales in that year). The key independent variables are merit (*R&D Efficiency*), corruption (*AETC/Sales*), the *Post Event* indicator (where the event is either the removal of a top provincial official on corruption charges or the departure of a technology bureau head), and interaction terms between merit or corruption and the *Post Event* indicator that equaled one for the three-year period after the event and zero otherwise. Control variables include an

indicator variable for SOEs, the firms' political connections, and performance and financial variables including *Return on Assets*, *Tobin's q*, and *Leverage*. (All control variables were lagged by one year. Detailed variable definitions are in Appendix B.)

Panel A of Table 5 reports the regression results pertaining to the removal of top provincial officials on corruption charges. Models (1) through (3) focus on the two key independent variables: *R&D Efficiency* and *AETC/Sales*. Both enter the regressions with positive and highly

Table 5. Panel Regressions: Merit, Corruption, and Subsidies

Dependent variable	(1) <i>Subsidies/Sales_{t+1}</i>	(2) <i>Subsidies/Sales_{t+1}</i>	(3) <i>Subsidies/Sales_{t+1}</i>	(4) <i>Subsidies/Sales_{t+1}</i>	(5) <i>Subsidies/Sales_{t+1}</i>	(6) <i>Subsidies/Sales_{t+1}</i>	(7) <i>Subsidies/Sales_{t+1}</i>
Panel A: Before and after the removal of top provincial officials on corruption charges							
<i>R&D Efficiency_t</i>	0.004*** (0.001)		0.005*** (0.001)	0.004*** (0.001)	0.003*** (0.001)	0.001 (0.001)	0.002 (0.003)
<i>AETC/Sales_t</i>		0.058** (0.023)	0.059*** (0.023)	0.078*** (0.026)	0.081*** (0.025)	0.062*** (0.020)	0.290** (0.128)
<i>Post Removal_t</i>			0.0002 (0.000)	−0.0002 (0.000)	−0.0001 (0.000)	0.0001 (0.000)	−0.00004 (0.000)
<i>R&D Efficiency_t × Post Removal_t</i>				0.014*** (0.005)	0.013*** (0.005)	0.006* (0.004)	0.009** (0.004)
<i>AETC/Sales_t × Post Removal_t</i>				−0.085*** (0.031)	−0.095*** (0.031)	−0.059*** (0.022)	−0.051* (0.030)
<i>SOE_t</i>					−0.0001 (0.000)	−0.001 (0.001)	−0.001 (0.001)
<i>Political Connection_t</i>					0.001* (0.000)	0.0001 (0.001)	0.0001 (0.001)
<i>Return on Assets_t</i>					0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
<i>Tobin's q_t</i>					−0.006*** (0.002)	−0.004*** (0.001)	−0.004*** (0.001)
<i>Leverage_t</i>					0.0001* (0.000)	0.0001 (0.000)	−0.0001 (0.000)
Constant	0.005*** (0.000)	0.005*** (0.000)	0.005*** (0.000)	0.005*** (0.000)	0.007*** (0.000)	0.006*** (0.000)	0.006*** (0.000)
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	No	No
Province fixed effects	Yes	Yes	Yes	Yes	Yes	No	No
Firm fixed effects	No	No	No	No	No	Yes	Yes
Province fixed effects × <i>R&D Efficiency_t</i>	No	No	No	No	No	No	Yes
Year fixed effects × <i>R&D Efficiency_t</i>	No	No	No	No	No	No	Yes
Province fixed effects × <i>AETC/Sales_t</i>	No	No	No	No	No	No	Yes
Year fixed effects × <i>AETC/Sales_t</i>	No	No	No	No	No	No	Yes
<i>N</i>	10,445	10,445	10,445	10,445	10,445	10,445	10,445
<i>R</i> ²	0.100	0.098	0.103	0.107	0.136	0.603	0.608
Panel B: Before and after unanticipated departures of provincial technology bureau heads							
<i>R&D Efficiency_t</i>	0.004*** (0.001)		0.005*** (0.001)	0.003*** (0.001)	0.002** (0.001)	0.0002 (0.001)	0.002 (0.003)
<i>AETC/Sales_t</i>		0.058** (0.023)	0.059*** (0.023)	0.082*** (0.027)	0.078*** (0.027)	0.077*** (0.022)	0.258** (0.128)
<i>Post Departure_t</i>			−0.0001 (0.000)	−0.001** (0.000)	−0.001** (0.000)	−0.0002 (0.000)	−0.0002 (0.000)
<i>R&D Efficiency_t × Post Departure_t</i>				0.006** (0.002)	0.006*** (0.002)	0.003** (0.001)	0.003** (0.001)
<i>AETC/Sales_t × Post Departure_t</i>				−0.069** (0.030)	−0.055* (0.031)	−0.081*** (0.023)	−0.079*** (0.026)
Constant	0.005*** (0.001)	0.005*** (0.001)	0.005*** (0.001)	0.005*** (0.001)	0.007*** (0.001)	0.006*** (0.001)	0.006*** (0.001)

Table 5. (Continued)

Panel B: Before and after unanticipated departures of provincial technology bureau heads									
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Lagged firm controls	No	No	No	No	No	No	No	No	No
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Province fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	No	No	No	No	No	No	No	No	No
Province fixed effects $\times$ $R\&D$ Efficiency _{<i>it</i>}	No	No	No	No	No	No	No	No	No
Year fixed effects $\times$ $R\&D$ Efficiency _{<i>it</i>}	No	No	No	No	No	No	No	No	No
Province fixed effects $\times$ AETC/Sales _{<i>it</i>}	No	No	No	No	No	No	No	No	No
Year fixed effects $\times$ AETC/Sales _{<i>it</i>}	No	No	No	No	No	No	No	No	No
N	10,445	10,445	10,445	10,445	10,445	10,445	10,445	10,445	10,445
R ²	0.100	0.098	0.103	0.106	0.135	0.135	0.135	0.603	0.608

Notes. This table reports results of panel regressions of R&D subsidies before and after (a) the removal of top provincial officials on corruption charges (Panel A) and (b) the unanticipated departures of provincial technology bureau heads (Panel B). The dependent variable is *Subsidies/Sales*, measured as the amount of R&D subsidies a firm received in a given year, divided by its annual revenue in that year. In Panel B, for brevity, only key coefficients are reported. Detailed variable definitions are found in Appendix B. Huber-White heteroskedasticity-consistent standard errors clustered by firm are used for all regressions. Standard errors are in parentheses.
*, **, ***Statistical significance at the 10%, 5%, and 1% levels, respectively.

significant signs, indicating that both were important determinants of subsidies. The coefficients in Panel A suggest that a one standard deviation increase in *R&D Efficiency* was associated with a 9.4% increase in R&D subsidies.¹⁶ Likewise, a one-standard-deviation increase in *AETC/Sales* was associated with an 8.1% increase in R&D subsidies.¹⁷ Thus, on average, corruption and merit had similar impacts on subsidies.

Models (4) through (7) include the interaction terms between *R&D Efficiency* and *AETC/Sales* and the *Post* indicators. In both Panels A and B and across all four specifications, we find that the interaction term between *R&D Efficiency* and the *Post* indicators was positive and significant, indicating that merit had a stronger effect on subsidies after either the removal of top provincial officials on corruption charges (Panel A) or after the unanticipated departure of technology bureau heads (Panel B). In contrast, the interaction term between *AETC/Sales* and the *Post* indicators was significantly negative across all four models in both panels, suggesting that corruption had a weaker effect on subsidies in postevent years.

Furthermore, in models (6) and (7), which include firm fixed effects, *R&D Efficiency* alone was not significant in predicting subsidies. Only the interaction term between *R&D Efficiency* and the *Post* indicators was positive and significant, an effect that appeared in both panels. Therefore, the positive effect of merit was concentrated in the postevent years when corruption had abated. On the other hand, *AETC/Sales* alone had a positive and significant coefficient, but its interaction term with the *Post* indicators had a negative and significant sign. Thus, corruption's contribution to subsidies was reduced in postevent years in both panels. With regard to control variables, we find that firms' political connections did not affect subsidies received; but profitability (*Return on Assets*) and *Leverage* were negatively associated with subsidies. This pattern seems consistent with the possibility that more profitable firms and those with ready access to credit had less need for subsidies.¹⁸

In Table 6, we repeat the analyses reported in Table 5, Panels A and B, column (4), adjusting them as follows to explore the heterogeneity of the effects:

- We interact a dummy variable denoting firms that have high (that is, above median) *R&D Efficiency* at the time of the observation with the continuous *R&D Efficiency* measure and with the interaction between the *R&D Efficiency* measure and the *Post Removal* (or *Post Departure*) dummy,
- We interact a dummy variable denoting firms that have a high (that is, above median) *AETC/Sales* ratio at the time of the observation with the continuous *AETC/Sales* ratio and with the interaction between the *AETC/Sales* ratio and the *Post Removal* (or *Post Departure*) dummy, or
- We undertake both steps simultaneously.

The results suggest that, although the effects regarding *R&D Efficiency* are present in both the high and low

Table 6. Cross-sectional Heterogeneity in the Main Effects

Dependent variable	(1) <i>Subsidies/Sales_{t+1}</i>	(2) <i>Subsidies/Sales_{t+1}</i>	(3) <i>Subsidies/Sales_{t+1}</i>
Panel A: Before and after the removal of top provincial officials on corruption charges			
<i>Dummy_Hi R&D Efficiency_t × R&D Efficiency_t × Post Removal_t</i>	0.014*** (0.005)		0.014*** (0.005)
<i>Dummy_Hi R&D Efficiency_t × R&D Efficiency_t</i>	0.004*** (0.001)		0.004*** (0.001)
<i>Dummy_Hi AETC/Sales_t × AETC/Sales_t × Post Removal_t</i>		−0.099*** (0.034)	−0.095*** (0.033)
<i>Dummy_Hi AETC/Sales_t × AETC/Sales_t</i>		0.072** (0.031)	0.073** (0.031)
<i>Post Removal_t</i>	−0.0002 (0.000)	0.0004 (0.000)	−0.0001 (0.000)
Panel B: Before and after unanticipated departures of provincial technology bureau heads			
<i>Dummy_Hi R&D Efficiency_t × R&D Efficiency_t × Post Departure_t</i>	0.006** (0.002)		0.006** (0.002)
<i>Dummy_Hi R&D Efficiency_t × R&D Efficiency_t</i>	0.003** (0.001)		0.003*** (0.001)
<i>Dummy_Hi AETC/Sales_t × AETC/Sales_t × Post Departure_t</i>		−0.077** (0.033)	−0.074** (0.032)
<i>Dummy_Hi AETC/Sales_t × AETC/Sales_t</i>		0.073** (0.031)	0.074** (0.031)
<i>Post Departure_t</i>	−0.0004** (0.000)	−0.00002 (0.000)	−0.0003 (0.000)

Notes. This table examines the heterogeneity in the results presented in Table 5. It reports the key coefficients from Table 5, Panel A, column (4) and Table 5, Panel B, column (4), (in column (1)) interacting a dummy variable denoting firms that have high (that is, above median) *R&D Efficiency* at the time of the observation with the continuous *R&D Efficiency* measure and with the interaction between the *R&D Efficiency* measure and the *Post Removal* (or *Post Departure*) dummy, (in column (2)) interacting a dummy variable denoting firms that have a high (that is, above median) *AETC/Sales* ratio at the time of the observation with the continuous *AETC/Sales* ratio and with the interaction between the *AETC/Sales* ratio and the *Post Removal* (or *Post Departure*) dummy, or (in column (3)) undertaking both steps simultaneously. Detailed variable definitions are found in Appendix B. Huber-White heteroskedasticity-consistent standard errors clustered by firm are used for all regressions. Standard errors are in parentheses.

*, **, *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

R&D Efficiency firms, the effect is stronger among the high *R&D Efficiency* firms. The negative effect of the *AETC/Sales* ratio on subsidies is entirely driven by the high *AETC/Sales* firms.

Overall, the regression results in Tables 5 and 6 are consistent with the DiD results in Figures 5 and 6 and our hypothesis. They indicate a shift in the weights of merit and corruption in determining R&D subsidies. Prior to both types of events, corruption had a significant effect, whereas merit had little effect; after both types of events, corruption's effect was negated, whereas merit had a significant impact.

One alternative explanation for our finding on the departure of technology bureau heads was that the new bureau heads were more qualified than their predecessors. To check this possibility, we use two variables to proxy for an official's qualifications: their working experience (the number of years an official had worked in the technology bureau system before becoming bureau head) and educational background (equal to three, two, one, or zero if the bureau head had a doctoral degree, master's degree, bachelor's degree, or none of these, respectively). The differences in the working experience and educational background between the new and old bureau heads were not statistically significant (t statistics = 0.105 and 0.785), suggesting that our results were unlikely to be driven by the higher ability of the new bureau heads. These results are reported in Table IA5 in the online appendix.

In summary, the results in this section indicate that both merit and corruption affected subsidies. However, corruption played a stronger role and merit played a weaker role when the overall level of corruption was higher. Allocations became more sensitive to merit and less to corruption after events that reduced chances of petty corruption.

4. Subsidies and Future Innovation

In this section, we examine the relationship between subsidies and future innovation outcomes, with a focus on how corruption affected this relationship. Results in the last section support our hypothesis that corruption rendered subsidy decisions less responsive to merit. Consequently, we expect that when corruption was rampant, the efficacy of subsidy programs was low and the relationship between subsidies and future innovation weak. In contrast, when corruption abated, this relationship should have been stronger. To examine this hypothesis, we use the same difference-in-differences methodology and estimate the following panel regression:

$$Y_{i,t+1} = \beta_0 Y_{i,t} + \beta_1 \text{Subsidies}/\text{Sales}_{i,t} + \beta_2 \text{Post Event}_{i,t} + \beta_3 \text{Subsidies}/\text{Sales}_{i,t} \times \text{Post Event}_{i,t} + \text{Controls} + \text{Fixed Effects}. \quad (4)$$

The dependent variables are the two innovation outcome measures defined in Section 2.2: *Patents/Sales* and

Relative Citation Strength. The key independent variables are the R&D subsidy in the prior year (scaled by sales), the *Post Event* indicator (where the observation is either the removal of a top provincial official on corruption charges or the departure of a technology bureau head), and the interaction term between the two. If corruption weakens the relationship between subsidies and future innovation, we expect the coefficient on the interaction term to be positive, since both events reduce the level of corruption. Control variables are the same as in the previous section.

Results for these analyses are reported in Tables 7 and 8. Panels A and B of each table examine *Patents/Sales* and *Relative Citation Strength*, respectively.

Table 7 examines the removal of top provincial officials on corruption charges. In Panel A, the coefficient on the key interaction term between subsidies and the *POST* indicator is positive and significant in all models, consistent with our hypothesis that the relationship between subsidies and future innovation was stronger when corruption had abated. The sign on the subsidies variable alone is positive in model (1) but became insignificant in models (2) to (6). The interaction term, however, remains highly significant, even with firm fixed effects. These results indicate that corruption had a material, negative impact on the relationship between subsidies and future innovation. In terms of magnitude, coefficients on the interaction term in models (2) to (4) imply that a one-standard-deviation increase in subsidies (as a percentage of sales) was associated with a 56% increase in the U.S. patenting rate in postevent years.¹⁹ Models (5) and (6) indicate that once firm fixed effects are added, subsidies were positively related to future subsidies only when corruption had abated. We find similar results in Panel B using the *Relative Citation Strength* as the dependent variable.

Table 8 presents a parallel set of results using the unanticipated departures of provincial technology bureau heads as the source of identification. Results in this table were consistent with those in Table 7. Notably, the relationship between subsidies and future innovation measured by either (scaled) patent counts or *Relative Citation Strength* was significantly stronger in postevent years than before.

In Table IA11 in the online appendix, we undertake several analyses to explore the robustness of these results. In Table IA11, Panel A, in the online appendix, we repeat the analysis reported in Table 7, Panel A, column (2), and in Table 8, Panel A, column (2), now winsorizing the upper end of dependent variable at the 1% through 5% level. The results in Tables 7 and 8 remain robust to this trimming. In Panel B, we repeat the analyses reported in Table 7, Panel A, column (2), and in Table 8, Panel A, column (2), now computing the dependent variable using Chinese rather than U.S. patents. In Panel C, we repeat the analyses reported in Table 7, Panel A, column (2), and in Table 8, Panel A, column (2), now using an unscaled

Table 7. Subsidies and Future Innovation: Before and After the Removal of Top Provincial Officials on Corruption Charges

Panel A: U.S. patents						
Dependent variable	(1) Patents/Sales (U.S.) _{t+1}	(2) Patents/Sales (U.S.) _{t+1}	(3) Patents/Sales (U.S.) _{t+1}	(4) Patents/Sales (U.S.) _{t+1}	(5) Patents/Sales (U.S.) _{t+1}	(6) Patents/Sales (U.S.) _{t+1}
<i>Subsidies/Sales_t</i>	0.003*** (0.001)	0.002 (0.001)	0.002 (0.001)	0.001 (0.001)	−0.001 (0.001)	−0.014 (0.010)
<i>Subsidies/Sales_t × Post Removal_t</i>		0.007** (0.003)	0.007** (0.003)	0.007** (0.003)	0.008** (0.004)	0.011** (0.005)
<i>Post Removal_t</i>		−0.00002 (0.000)	−0.00002 (0.000)	−0.00001 (0.000)	−0.00002 (0.000)	−0.00004* (0.000)
<i>Patents/Sales (U.S.)_t</i>	0.614*** (0.063)	0.613*** (0.063)	0.611*** (0.063)	0.602*** (0.063)		
<i>Ln(Assets)_t</i>			0.00001 (0.000)	0.00001* (0.000)	−0.0001** (0.000)	−0.0001** (0.000)
<i>Ln(Age)_t</i>			0.00002 (0.000)	0.00002 (0.000)	−0.0001 (0.000)	−0.0001 (0.000)
<i>Leverage_t</i>			−0.00004** (0.000)	−0.00003 (0.000)	−0.0002*** (0.000)	−0.0001*** (0.000)
<i>Intangible Assets/Assets_t</i>			−0.00001 (0.000)	−0.0001 (0.000)	−0.0003 (0.000)	−0.0003 (0.000)
<i>Return on Assets_t</i>			0.0001 (0.000)	0.0001* (0.000)	−0.0001 (0.000)	−0.0001 (0.000)
<i>Tobin's q_t</i>			0.000001 (0.000)	0.000002 (0.000)	−0.00001** (0.000)	−0.00001** (0.000)
<i>SOE_t</i>			−0.00001 (0.000)	−0.00001 (0.000)	0.00001 (0.000)	0.000001 (0.000)
<i>Political Connection_t</i>			0.00002 (0.000)	0.00002 (0.000)	0.00002 (0.000)	0.00002 (0.000)
Constant	0.00002*** (0.000)	0.00002*** (0.000)	−0.00002 (0.000)	−0.00002* (0.000)	0.001*** (0.000)	0.001*** (0.000)
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	No	No	No	Yes	No	No
Province fixed effects	No	No	No	Yes	No	No
Firm fixed effects	No	No	No	No	Yes	Yes
Province fixed effects × <i>Subsidies/Sales_t</i>	No	No	No	No	No	Yes
Year fixed effects × <i>Subsidies/Sales_t</i>	No	No	No	No	No	Yes
N	10,696	10,696	10,696	10,696	10,696	10,696
R ²	0.426	0.428	0.429	0.433	0.611	0.616

Table 7. (Continued)

Panel B: U.S. relative patent citation strength						
Dependent variable	(1) Relative Citation Strength (U.S.) _{t+1}	(2) Relative Citation Strength (U.S.) _{t+1}	(3) Relative Citation Strength (U.S.) _{t+1}	(4) Relative Citation Strength (U.S.) _{t+1}	(5) Relative Citation Strength (U.S.) _{t+1}	(6) Relative Citation Strength (U.S.) _{t+1}
<i>Subsidies/Sales_t</i>	1.219** (0.592)	0.744 (0.504)	0.848 (0.542)	0.666 (0.519)	−0.486 (0.432)	−2.442 (1.799)
<i>Subsidies/Sales_t × Post Removal_t</i>		2.207* (1.267)	2.243* (1.271)	2.095* (1.235)	2.355* (1.399)	4.750** (2.244)
<i>Post Removal_t</i>		−0.012* (0.007)	−0.012* (0.007)	−0.008 (0.007)	−0.009 (0.008)	−0.019* (0.010)
<i>Relative Citation Strength (U.S.)_t</i>	0.370*** (0.045)	0.369*** (0.045)	0.359*** (0.044)	0.341*** (0.044)		
Constant	0.017*** (0.003)	0.020*** (0.003)	−0.371*** (0.078)	−0.426*** (0.081)	0.102 (0.152)	0.107 (0.153)
Lagged firm controls	No	No	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	No	No	No	Yes	No	No
Province fixed effects	No	No	No	Yes	No	No
Firm fixed effects	No	No	No	No	Yes	Yes
Province fixed effects × <i>Subsidies/Sales_t</i>	No	No	No	No	No	Yes
Year fixed effects × <i>Subsidies/Sales_t</i>	No	No	No	No	No	Yes
N	10,696	10,696	10,696	10,696	10,696	10,696
R ²	0.149	0.150	0.157	0.169	0.401	0.406

Notes. This table investigates the relationship between R&D subsidies and future innovation before and after the removal of top provincial officials on corruption charges. We use two measures of future innovation as the dependent variable. In Panel A, the dependent variable is *Patents/Sales* using U.S. patent data. In Panel B, the dependent variable is *Relative Citation Strength* using U.S. patent data. In Panels B, for brevity, only key coefficients are reported. Detailed variable definitions are found in Appendix B. In columns (1) through (4), we include the lagged dependent variable to control for persistence. In columns (5) and (6), we exclude the lagged dependent variable when we include firm fixed effects. Huber-White heteroskedasticity-consistent standard errors clustered by firm are used for all regressions. Standard errors are in parentheses.

*, **, ***Statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 8. Subsidies and Future Innovation: Before and After the Unanticipated Departures of Provincial Technology Bureau Heads

Dependent variable	Panel A: U.S. patents					
	(1)	(2)	(3)	(4)	(5)	(6)
	<i>Patents/Sales</i> (U.S.) _{t+1}	<i>Patents/Sales</i> (U.S.) _{t+1}	<i>Patents/Sales</i> (U.S.) _{t+1}	<i>Patents/Sales</i> (U.S.) _{t+1}	<i>Patents/Sales</i> (U.S.) _{t+1}	<i>Patents/Sales</i> (U.S.) _{t+1}
<i>Subsidies/Sales_t</i>	0.003*** (0.001)	0.001* (0.001)	0.001 (0.001)	0.001 (0.001)	−0.002* (0.001)	−0.011 (0.011)
<i>Subsidies/Sales_t × Post Departure_t</i>		0.004** (0.002)	0.004** (0.002)	0.004** (0.002)	0.006** (0.003)	0.006* (0.003)
<i>Post Departure_t</i>		−0.00002** (0.000)	−0.00002** (0.000)	−0.00002 (0.000)	−0.00001 (0.000)	−0.00002 (0.000)
<i>Patents/Sales (U.S.)_t</i>	0.614*** (0.063)	0.612*** (0.063)	0.610*** (0.063)	0.601*** (0.063)		
<i>Ln(Assets)_t</i>			0.00001 (0.000)	0.00001* (0.000)	−0.0001* (0.000)	−0.0001** (0.000)
<i>Ln(Age)_t</i>			0.00002 (0.000)	0.00002 (0.000)	−0.0001 (0.000)	−0.0001 (0.000)
<i>Leverage_t</i>			−0.00004** (0.000)	−0.00004* (0.000)	−0.0001*** (0.000)	−0.0001*** (0.000)
<i>Intangible Assets/Assets_t</i>			−0.00001 (0.000)	−0.0001 (0.000)	−0.0003 (0.000)	−0.0003 (0.000)
<i>Return on Assets_t</i>			0.0001 (0.000)	0.0001 (0.000)	−0.0001 (0.000)	−0.0001 (0.000)
<i>Tobin's q_t</i>			0.000001 (0.000)	0.000002 (0.000)	−0.00001** (0.000)	−0.00001** (0.000)
<i>SOE_t</i>			−0.00001 (0.000)	−0.00001 (0.000)	0.00001 (0.000)	0.00002 (0.000)
<i>Political Connection_t</i>			0.00002 (0.000)	0.00002 (0.000)	0.00003 (0.000)	0.00003 (0.000)
Constant	0.00002*** (0.000)	0.00002*** (0.000)	−0.00002 (0.000)	−0.00002* (0.000)	0.001*** (0.000)	0.001*** (0.000)
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	No	No	No	Yes	No	No
Province fixed effects	No	No	No	Yes	No	No
Firm fixed effects	No	No	No	No	Yes	Yes
Province fixed effects × <i>Subsidies/Sales_t</i>	No	No	No	No	No	Yes
Year fixed effects × <i>Subsidies/Sales_t</i>	No	No	No	No	No	No
N	10,696	10,696	10,696	10,696	10,696	10,696
R ²	0.426	0.427	0.428	0.432	0.611	0.615

Table 8. (Continued)

Dependent variable	Panel B. U.S. relative patent citation strength					
	(1)	(2)	(3)	(4)	(5)	(6)
	Relative Citation Strength (U.S.) _{t+1}	Relative Citation Strength (U.S.) _{t+1}	Relative Citation Strength (U.S.) _{t+1}	Relative Citation Strength (U.S.) _{t+1}	Relative Citation Strength (U.S.) _{t+1}	Relative Citation Strength (U.S.) _{t+1}
<i>Subsidies/Sales_t</i>	1.219** (0.592)	0.322 (0.366)	0.428 (0.409)	0.261 (0.413)	−0.698 (0.468)	−1.647 (1.874)
<i>Subsidies/Sales_t × Post Departure_t</i>		2.057** (1.003)	2.034** (0.975)	1.917** (0.949)	1.528* (0.841)	1.551* (0.904)
<i>Post Departure_t</i>		−0.006 (0.006)	−0.004 (0.006)	−0.005 (0.007)	−0.001 (0.006)	−0.001 (0.007)
<i>Relative Citation Strength (U.S.)_t</i>	0.370*** (0.045)	0.368*** (0.045)	0.358*** (0.044)	0.341*** (0.044)		
Constant	0.017*** (0.003)	0.020*** (0.003)	−0.375*** (0.077)	−0.426*** (0.080)	0.082 (0.150)	0.104 (0.154)
Lagged firm controls	No	No	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	No	No	No	Yes	No	No
Province fixed effects	No	No	No	Yes	No	No
Firm fixed effects	No	No	No	No	Yes	Yes
Province fixed effects × <i>Subsidies/Sales_t</i>	No	No	No	No	No	Yes
Year fixed effects × <i>Subsidies/Sales_t</i>	No	No	No	No	No	Yes
<i>N</i>	10,696	10,696	10,696	10,696	10,696	10,696
<i>R</i> ²	0.149	0.150	0.157	0.166	0.401	0.404

Notes. This table investigates the relationship between R&D subsidies and future innovation around the unanticipated departures of provincial technology bureau heads. We use two measures of future innovation as the dependent variable. In Panel A, the dependent variable is *Patents/Sales* using U.S. patent data. In Panel B, the dependent variable is *Relative Citation Strength* using U.S. patent citation data. In Panels B, for brevity, only key coefficients are reported. All variables are defined as in Appendix B. In columns (1) to (4), we include the lagged dependent variable to control for persistence. In columns (5) and (6), we exclude the lagged dependent variable when we include firm fixed effects. Huber-White heteroskedasticity-consistent standard errors clustered by firm are used for all regressions. Standard errors are in parentheses.

*, **, ***Statistical significance at the 10%, 5%, and 1% levels, respectively.

dependent variable. We also use sales in period t as a separate dependent variable and use either a Poisson or negative binomial specification. Across the two panels, the results are generally robust. (The one exception is when we use the negative binomial specification and the unanticipated turnover of technology bureau heads.) The robustness of these results (as well as other unreported ones, such as using the Chinese data with winsorization) suggests that the findings are not sensitive to a particular specification.

One concern is that some of these patterns may have started before the removal of the corrupt official or the departure of the bureau head. To check for pretrends in the subsidies-innovation relationship, we estimate a dynamic differences-in-differences model. As reported in Table IA6 in the online appendix, we fail to find significant pretrends in the subsidies-innovation relationship before the removal of corrupt officials and the departure of technology bureau heads.

Consistent with our hypothesis, the results in this section indicate that the relationship between subsidies and future innovation became significantly stronger after both types of events. We emphasize that these results do not indicate that subsidies caused firm-level innovation. Instead, the consistently stronger relationship between subsidies and future innovation after corruption-reducing events suggests that corruption negatively affected the efficacy of subsidy programs.

5. Alternative Sources of Financing

The results in the previous two sections show that corruption negatively affected the quality of subsidy-

granting decisions. One key question is whether these decisions ultimately mattered for economic outcomes. If innovative firms could tap alternative sources of capital, they might have circumvented the allocative inefficiencies of subsidies. Given that 20% of Chinese firms' R&D funds came from subsidies, it might appear that the allocation efficiency per se was important. Nevertheless, we investigate this question by examining firms' dependence on external financing, the availability of alternative sources of capital, and firms' actual subsequent fundraising patterns.

To partition firms based on their reliance on external financing, we construct an industry-level measure based on Rajan and Zingales (1998) (RZ). First, for each firm, we calculate its average external-financing dependence using data from 2009, 2010, and 2011, the three years before the anticorruption campaign. A firm's dependence on external financing in a given year is measured as its capital expenditures in that year minus its cash flow from operations divided by capital expenditures. We then calculate the equal-weighted average of this financing dependence measure across all firms in a three-digit CSRC industry. We used the median of the industry-level financial dependence to divide all industries into those that have high or low dependence on external finance.²⁰

To proxy for the availability of alternative financing, we construct measures of bank and venture capital availability at the province level. Using data from 2009, 2010 and 2011, we calculate the average bank's total assets in a province (scaled by provincial GDP) and the number of venture firms present in a province (also scaled by provincial GDP).²¹ We classify provinces where either of

Table 9. Subsidies and External Financing

Dependent variable	Dependence on external financing		Bank and VC capital availability	
	(1) High <i>Subsidies/Sales_{t+1}</i>	(2) Low <i>Subsidies/Sales_{t+1}</i>	(3) High <i>Subsidies/Sales_{t+1}</i>	(4) Low <i>Subsidies/Sales_{t+1}</i>
Panel A: Before and after the removal of top provincial officials on corruption charges				
<i>R&D Efficiency_t × Post Removal_t</i>	0.009*** (0.003)	0.007* (0.004)	0.004 (0.007)	0.009** (0.004)
<i>AETC/Sales_t × Post Removal_t</i>	−0.117** (0.059)	−0.010 (0.039)	−0.140* (0.073)	−0.013 (0.034)
Panel B: Before and after unanticipated departures of provincial technology bureau heads				
<i>R&D Efficiency_t × Post Departure_t</i>	0.005** (0.002)	0.002 (0.002)	0.001 (0.002)	0.005*** (0.002)
<i>AETC/Sales_t × Post Departure_t</i>	−0.126*** (0.043)	−0.042 (0.037)	−0.130** (0.063)	−0.072** (0.031)

Notes. This table examines the allocation of R&D subsidies to firms in industries with high or low dependence on external financing (columns (1) and (2)) and in provinces with high or low bank and venture capital availability (columns (3) and (4)). The dependent variable is *Subsidies/Sales_{t+1}*, measured as the amount of R&D subsidies a firm received in a given year, divided by its revenue in that year. The empirical specification is identical to model (7) in Table 5. We classify a firm as having high (low) dependence on external financing if its industry RZ index is above (below) median in the years 2009 to 2011. We classify a province as having high (low) bank and venture capital availability if the average total assets of banks in the province (scaled by provincial GDP) or the number of venture capital firms present in the province (also scaled by provincial GDP) is above the 75th percentile over the period of 2009 to 2011. For brevity, only key coefficients are reported. Huber-White heteroskedasticity-consistent standard errors clustered by firm are used for all regressions. Standard errors are in parentheses.

*, **, ***Statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 10. Subsidies, External Financing, and Future Innovation

Dependent variable	Dependence on external financing				Bank and VC capital availability			
	(1) High	(2) Low	(3) High	(4) Low	(5) High	(6) Low	(7) High	(8) Low
	<i>Patents/Sales</i> <i>(U.S.)_{t+1}</i>		<i>Relative Citation</i> <i>Strength (U.S.)_{t+1}</i>		<i>Patents/Sales (U.S.)_{t+1}</i>		<i>Relative Citation</i> <i>Strength (U.S.)_{t+1}</i>	
Panel A: Before and after the removal of top provincial officials on corruption charges								
<i>Subsidies/Sales_t × Post Removal_t</i>	0.017** (0.007)	0.002 (0.002)	8.048*** (2.851)	−0.118 (0.640)	0.005 (0.005)	0.012* (0.006)	0.674 (1.330)	5.924** (2.731)
Panel B: Before and after unanticipated departures of provincial technology bureau heads								
<i>Subsidies/Sales_t × Post Departure_t</i>	0.012** (0.005)	−0.002 (0.002)	2.892* (1.528)	−0.082 (0.549)	0.009 (0.006)	0.004 (0.004)	0.714 (1.091)	1.923 (1.301)

Notes. This table investigates the relationship between R&D subsidies and future innovation for firms with high or low dependence on external financing (columns (1) to (4)) and in provinces with high or low bank or venture capital availability (columns (5) to (8)). The regression model is identical to model (6) in Tables 7 and 8. We classify a firm as having high (low) dependence on external financing if its industry *RZ* index is above (below) median in the years 2009 to 2011. We classify a province as having high (low) bank and venture capital availability if the average total assets of banks in the province (scaled by provincial GDP) or the number of venture capital firms present in the province (also scaled by provincial GDP) is above the 75th percentile over the period of 2009 to 2011. For brevity, only key coefficients are reported. Huber-White heteroskedasticity-consistent standard errors are clustered by firm and used for all regressions. Standard errors are in parentheses.

*, **, ***Statistical significance at the 10%, 5%, and 1% levels, respectively.

these measures was above the 75th percentile as having high capital availability and the rest as having low capital availability.

Table 9 re-examines the determinants of subsidies in subsamples partitioned by dependence on external financing and by capital availability. For brevity, we report only the regression results using model (7) in Table 5, that is, the specification that includes firm fixed effects. Our main results—the increased impact of merit and the decreased impact of corruption on subsidy allocation—are concentrated in industries that relied heavily on external financing and in provinces that had low capital availability.

Table 10 re-examines the subsidy-future innovation relationship, focusing on the same two partitions of the firms. For brevity, we report only the regression results using model (6) in Tables 7 and 8, that is, the specification that includes firm fixed effects. Our baseline result of a stronger connection between subsidies and future

innovation after corruption-abating events is concentrated in industries that relied heavily on external financing and in provinces with low capital availability.

Taken together, the results in Tables 9 and 10 suggest that alternative sources of financing were unlikely to be a substitute for subsidies for many firms, particularly those based in regions where alternative sources of capital were scarce and those in industries that relied heavily on external financing. These results highlight the cost of corruption and the resulting allocational inefficiency of subsidies.

6. Additional Analyses

6.1. Deterrence Effect of the Anticorruption Campaign

One of our identification methods relies on the removal of a top provincial official on corruption charges. We

Table 11. Deterrence Effect

Dependent variable	Whole sample (1)	Corruption index	
	<i>Subsidies/Sales_{t+1}</i>	(2) High <i>Subsidies/Sales_{t+1}</i>	(3) Low <i>Subsidies/Sales_{t+1}</i>
<i>R&D Efficiency_t × Post_2012_t</i>	0.002 (0.002)	0.003 (0.004)	0.002 (0.003)
<i>AETC/Sales_t × Post_2012_t</i>	−0.144*** (0.048)	−0.182** (0.089)	−0.141** (0.056)

Notes. This table examines the deterrence effect of the anticorruption campaign. We examine subsidy granting decisions by repeating the regression in model (7) of Table 5 for various subsamples. We use 2012 (the start of the anticorruption campaign) as the cutoff year and focus on subsidy decisions made in the three years before (2009 to 2011) versus three years after (2013 to 2015). We estimate the regression for the whole sample (column (1)), as well as for subsamples sorted by an ex ante measure of provincial level corruption (columns (2) and (3)). Details of the corruption index are found in Section 6.1. For brevity, only key coefficients are reported. Huber-White heteroskedasticity-consistent standard errors are clustered by firm and used for all regressions. Standard errors are in parentheses.

*, **, ***Statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 12. Social Network Effect

Dependent variable	(1) Top provincial officials <i>Subsidies/Sales_{t+1}</i>	(2) Technology bureau heads <i>Subsidies/Sales_{t+1}</i>
<i>R&D Efficiency_t × Post Rotation_t</i>	−0.002 (0.002)	0.004*** (0.002)
<i>AETC/Sales_t × Post Rotation_t</i>	−0.035* (0.019)	−0.057** (0.025)

Notes. This table examines the social network effect by examining subsidy decisions before and after in-province rotations of top provincial officials and provincial technology bureau heads. For brevity, only key coefficients are reported. Huber-White heteroskedasticity-consistent standard errors are clustered by firm are used for all regressions. Standard errors are in parentheses.

*, **, *** indicates statistical significance at the 10%, 5%, and 1% levels, respectively.

find that after such events, subsidy decisions were less sensitive to corruption but more sensitive to merit; hence, the efficacy of subsidy decisions improved. One interesting question is whether the ex ante deterrent effect of anticipated anticorruption actions rather than the ex post punishment of corruption in a given province would suffice to bring about the same type of result.

To test this hypothesis, we use 2012—the onset of the anticorruption campaign but before any major removals of officials have been carried out—as the “shock” year. We examine subsidy decisions in the three years before (2009 to 2011) and three years after (2013 to 2015).²² This test is by no means perfect, as the one-time shock may have been concurrent with other events happening in China. Nevertheless, the results reported in Table 11 provide some evidence that the threat of the anticorruption campaign was effective in reducing the sensitivity of subsidy decisions on corruption. The coefficient in the regression of subsidies on the interaction term between *AETC/Sales* and the *Post_2012* indicator is negative and significant in the whole sample (column (1)), as well as in both subsamples sorted by an ex ante measure of provincial-level corruption (columns (2) and (3)).²³ This result

suggests a deterrence effect. However, the threat of the anticorruption campaign did not increase the sensitivity of subsidy decisions to firm merit in any of the sample splits, unlike in our baseline results. Thus, the threat of the anticorruption campaign seemed to have a deterrent effect on corruption but did not lead to more merit-based decisions.

6.2. Social Connections and Career Incentives

Our second identification method relies on the departures of technology bureau heads. After these events, subsidy decisions were less sensitive to corruption and more sensitive to firm merit. This finding raises the broader question of whether the results stemmed from the disruption of social connections between officials and firms. If so, any event that disrupted these connections might have had the same effect.

To examine whether disruption of social connections is necessary for the improvement of subsidy decisions, we collect data on routine (noncorruption related) rotations of provincial top officials (provincial governors and party secretaries)²⁴ and new technology bureau heads from 2007 to 2018, focusing on cases where the newly

Table 13. Subsidy Granting Decisions by Subsidy Types

Dependent variable	<i>Subsidies/Sales_{t+1}</i>	
	Subsidies strongly related to innovation	Subsidies weakly related to innovation
Panel A: Before and after the removal of top provincial officials on corruption charges		
<i>R&D Efficiency_t × Post Removal_t</i>	0.008** (0.003)	0.001 (0.001)
<i>AETC/Sales_t × Post Removal_t</i>	−0.042* (0.025)	0.009 (0.015)
Panel B: Before and after unanticipated departures of provincial technology bureau heads		
<i>R&D Efficiency_t × Post Departure_t</i>	0.003** (0.001)	−0.0003 (0.001)
<i>AETC/Sales_t × Post Departure_t</i>	−0.069*** (0.022)	−0.014 (0.011)

Notes. This table is a robustness check regarding Table 5. We re-examined the subsidy allocations by separating the seven subsidy categories into those that are strongly related to innovation (categories 1, 2, 3, 4, and 6) and those that are weakly related to innovation (categories 5 and 7). Details of the subsidy categories are found in Table 2. The regression specification in this table is identical to model (7) in Table 5. For brevity, only key coefficients are reported. Huber-White heteroskedasticity-consistent standard errors clustered by firm are used for all regressions. Standard errors are in parentheses.

*, **, ***Statistical significance at the 10%, 5%, and 1% level, respectively.

Table 14. Subsidies and Future Innovation by Subsidy Type

Dependent variable	Patents/Sales (U.S.) _{t+1}		Relative Citation Strength (U.S.) _{t+1}	
	Subsidies strongly related to innovation	Subsidies weakly related to innovation	Subsidies strongly related to innovation	Subsidies weakly related to innovation
Panel A: Before and after the removal of top provincial officials on corruption charges				
<i>Subsidies/Sales_t × Post Removal_t</i>	0.011*	0.012*	5.655**	2.714
	(0.006)	(0.006)	(2.677)	(2.160)
Panel B: Before and after unanticipated departures of provincial technology bureau heads				
<i>Subsidies/Sales_t × Post Departure_t</i>	0.007*	0.005	2.125*	1.019
	(0.004)	(0.005)	(1.125)	(1.633)

Notes. This table is a robustness check regarding Tables 7 and 8. We re-examine the relationship between subsidies and future innovation by separating the seven subsidy categories into those that are strongly related to innovation (categories 1, 2, 3, 4, and 6) and those that are weakly related to innovation (categories 5 and 7). Details of the subsidy categories are found in Table 2. Panel A examines the relationship between subsidies and future innovation before and after the removal of top provincial officials (parallel to Table 7) and Panel B focuses on before and after the unanticipated departure of provincial technology bureau heads (parallel to Table 8). The regression specification is identical to model (6) in Tables 7 and 8. For brevity only key coefficients are reported. Huber-White heteroskedasticity-consistent standard errors clustered by firm are used for all regressions. Standard errors are in parentheses.

*, **, ***Statistical significance at the 10%, 5%, and 1% level, respectively.

appointed officials were promoted from within the province. In these cases, the social connections between officials and firms were not disrupted. If disruption of social connection was the critical factor, we should not observe the types of changes seen in our baseline results. Table 12 shows that, even in these cases, subsidy decisions became less sensitive to corruption and more sensitive to merit after the rotations. These results are stronger in the rotations involving technology bureau heads. We interpret our results as indicating that disruption of the social connections between the relevant officials and firms was not necessary for the effects that we document. Newly appointed officials might have had strong enough career incentives to improve decision making.

Results in these two sections indicate that neither punishment for corruption nor disruption of social networks was necessary to effect better subsidy decisions. Anticorruption efforts had some deterrence effects to nonprosecuted officials. Meanwhile, newly appointed officials, even those who maintained social connections with the firms, made better decisions. In both cases, one reasonable interpretation was that officials’ career concerns played a role. Both types of events not only reduced the incentive or opportunity for corruption but also increased the incentive for good decisions.

6.3. Robustness Checks

We perform several robustness tests in addition to those mentioned in the previous sections. There were seven types of subsidies in China (listed in Table 2). Among them, categories 5 and 7 (subsidies for commercialization and equipment purchases and patent application fees) might have been less central to innovation than others. We expect the results to be stronger for subsidies

that were more directly related to innovation. Results reported in Tables 13 and 14 confirm this conjecture.

We also conduct a propensity-score matched sample analysis to confirm our main results. Specifically, we match each treated firm with one control firm with the most similar firm characteristics (using firm size, age, leverage, ROA, Tobin’s q, percentage of tangible assets, SOE status, political connections, and industry). The results using the matched sample (reported in Tables IA9 and IA10 in the online appendix) are qualitatively similar to those reported earlier.

7. Conclusions

Governments all over the world subsidize innovation efforts. Individuals allocating resources (R&D subsidies) can be susceptible to corruption, which might lead to resource misallocation. Using data from China, we investigated how corruption affected the relationship between government R&D subsidies and firm innovation. China was a fertile testing ground for these issues, given the pervasiveness of innovative subsidies and corruption, as well as the recent anticorruption efforts.

Exploiting the replacements of provincial political leaders as part of the anticorruption campaign and unanticipated changes in government officials responsible for provincial innovation programs as two identification methods, we established several main findings. First, firms’ innovative abilities and corruption-related expenditure activities affected the allocation of government R&D subsidies. Subsidies became significantly more sensitive to merit and less sensitive to corruption after both types of turnover. Second, after both types of events, the association between R&D subsidies and future innovation became significantly positive, whereas

they were weak beforehand. Both effects were stronger among firms in industries with greater dependence on external finance and in regions with fewer banks and venture capitalists. When we attempted to unpack the mechanisms of these results, our evidence suggests that they stem from both reduced incentives and opportunities for corruption and career concerns.

A number of questions were unanswered by our study. For example, although we focused on the overall efficiency of subsidy allocations, we were unable to address the effectiveness of particular program design elements, which can be critical for policy success (Howell 2017). Similarly, the generalizability of the results beyond the particular institutional setting remains unresolved. We leave such important questions for future research.

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Appendix A. Patent-Related Measures

All patent-related data used in this paper are obtained from InnoJoy, a subscription-based commercial database providing patent-related data on Chinese firms. InnoJoy covers not only patent filings in China, but also the United States and Patent Cooperation Treaty (PCT; administered by the World Intellectual Property Organization) applications and awards for Chinese firms.

In this paper, we primarily use Chinese invention patent data and U.S. utility patent data. As discussed in Endnotes 9 and 11 in the paper, both patent types are the most innovative kinds in their respective patent offices.

We use Chinese invention patent counts to calculate a firm's R&D efficiency according to Equation (2) in the paper. Our definition of the patent count for a given firm i in a given year t is the count of invention patents filed by firm i with the Chinese State Intellectual Property Office in year t that were approved by August 31, 2021. (Trailing) R&D efficiency is a key factor for subsidy grant decisions. For Chinese officials making those subsidy decisions, we deem it more likely that they focused on Chinese domestic patents. Therefore, we construct a variable that is more likely to reflect their information set.

We use a firm's U.S. utility patent counts (scaled by sales) and their *Relative Citation Strength* to measure a firm's future innovation. U.S. patents are generally regarded as much harder to obtain than Chinese ones and higher in innovation

content. Therefore, we use U.S. patent-related measures to measure a firm's future innovations. We count a firm's U.S. patents in the same way we count its Chinese patents: the patent count for a given firm i in a given year t is the count of utility patents filed by firm i with the U.S. Patent and Trademark Office in year t that were granted by August 31, 2021. *Relative Citation Strength* is similarly calculated based on the relevant patents filed by a firm in a given year, according to our description in Section 2.2 of the paper and the definitions in Appendix B.

We use a search and merge process to combine patent data from InnoJoy with firms' financial and other data. To obtain firm-level patent information from InnoJoy, we search the Chinese name of all Chinese listed firms and their wholly owned subsidiaries in the InnoJoy database. We then merge the search results with firms' financial and other information by firm name. If a firm's name changed over time, we collect all the historical names of the firm from the WIND database. We obtain information on wholly owned subsidiaries from firms' annual reports.

A similar search and merge process is followed to obtain firms' U.S. patent information. Specifically, to obtain the U.S. patent data from InnoJoy for a given firm, we search the firm's English trade name in the InnoJoy database and restrict the search to U.S. utility patents. When searching using English trade names of firms in InnoJoy, we drop suffixes such as "Co., Ltd.," "Company Limited," "Inc.," and so forth from the names to avoid matching difficulties. We also collect all the historic English name changes of the listed Chinese firms from the WIND database to construct a complete patent history in cases of name changes. Information on wholly owned subsidiaries was again obtained from firms' annual reports.

In the paper, virtually all our results are based on patents that were applied for and granted by August 31, 2021; that is, they are based on successful patent applications. We adjusted the counts of patents applied for in 2015 to 2018 upward to address the likelihood that additional patents filed in those years would ultimately be issued. We used here the time-adjusted methodology of Hall et al (2005) and Lerner and Seru (2022), using the grant-lag distributions of mature patents filed by Chinese firms to compute the adjustment factors.

In robustness checks, we also re-estimate our main regressions using total Chinese patent application counts, regardless of the approval outcome of the applications. These unreported results are robust to our main results and are available upon request.

Appendix B. Variable Definitions

AETC/Sales: AETC stands for firms' Abnormal Entertainment and Travel Costs scaled by firms' annual revenue and is calculated as the residual of Equation (1).

Age: The number of years between a company's establishment and the year of the observation.

Business in Other Regions: The number of geographical regions outside its home region from which a firm derives revenue. Chinese provinces are grouped into eight geographic regions: North (华北), South (华南), Middle (华中), East (华东), North-East (东北), North-West (西北), South-West

(西南), and Hong Kong/Macau/Taiwan (港, 澳, 台). Foreign countries are coded as one separate region.

ETC/Sales: ETC stands for firms' Entertainment and Travel Costs, reported in firm's annual statements. *ETC/Sales* is a firm's amount of ETC spending, divided by its annual revenue.

Intangible Assets/Assets: The book value of a firm's intangible assets divided by the book value of its total assets.

Leverage: A firm's book value of total liabilities divided by its book value of total assets.

Ln(Age): The natural logarithm of the number of years between a company's establishment and the year of the observation.

Ln(Assets): The natural logarithm of a firm's book value of total assets (in RMB).

Patents/Sales (U.S.): The number of patent applications filed by a firm in the United States in a given year that are ultimately granted by August 31, 2021, divided by its revenue (measured in millions of RMB) in that year.

PerCapGDP: The log of per capita GDP (in RMB) of the firm's home province.

Political Connection: An indicator variable that equals one if a company's CEO is currently or formerly an officer in the central government, or a local government, or the military and zero otherwise (as in Fan et al. (2007)). Data on CEOs are manually collected from the "Profile of Directors and Senior Managers" section of the firms' IPO prospectuses and annual reports.

Post Departure: An indicator variable that equals one for the three-year period after a firm's headquarter province has seen the unanticipated departure of provincial technology bureau officials from his/her post.

Post Removal: An indicator variable that equals one for the three-year period after a firm's headquarter province has seen the removal of top provincial officials for corruption charges.

Relative Citation Strength (U.S.): The number of patent citations that received up to August 31, 2021, per U.S. patent applied for in year t by firm i (that was ultimately granted by August 31, 2021), divided by the number of citations per U.S. patent received over the same period by all Chinese patents applied for in year t (that was also ultimately granted by August 31, 2021) in the same four-digit technological class.

Return on Assets (ROA): A firm's net income in a given year divided by its book value of total assets in the same year.

R&D Efficiency: A firm's R&D efficiency, calculated as Equation (2) in the paper as the ratio of the number of Chinese patents applied by a firm in a given year that were ultimately approved, divided by a capitalized measure of R&D expenditure measured in millions of RMB.

R&D/Sales: A firm's R&D expenditure in a year divided by its revenues in the same year.

RZ (Rajan-Zingales) measure of dependence on external financing: The average across all the sample firms in the firm's industry (using the Chinese Securities Regulatory Commission's three-digit industry classification codes) of the ratio of capital expenditures in a given year minus cash flow from operations in the same year, divided by capital expenditure.

Size (Mil. RMB): The book value of a firm's total assets in a given year, in millions of RMB.

SOE: An indicator variable if a company's largest ultimate shareholder, as disclosed in its annual statements, is a government entity. We classify firms' ownership types by tracing their ultimate ownership identified through annual statements.

Subsidies/Sales: The amount of R&D subsidies a firm receives in a given year, divided by its annual revenue in that year.

Tobin's q : A firm's market value of equity (average share price in a given year multiplied by its average number of shares outstanding) plus its book value of debt, divided by the book value of its total assets.

Endnotes

¹ Gong and Wu (2012) estimate that corruption cases involving mid-level officials accepting bribes, gifts, and entertainment accounted for about two-thirds of the total government official corruption cases from 2000 to 2009.

² We search for the key word "anti-corruption" in the titles of all newspaper articles published in all official provincial government newspapers between 2007 and 2018. In China, each provincial-level government has an official publication called the "Daily": for example, Beijing Daily, Beijing, Shanghai, Tianjin, and Chongqing are four municipalities that enjoy the same administrative status as a province.

³ See <http://www.ccdi.gov.cn/special/zyxszt/> (accessed February 10, 2019). For an excellent discussion of the central role of the CCDI in China's anti-corruption campaign, see Chen and Kung (2019). In China, the prosecution of government officials started within the CCDI. The cases were then brought to a court hearing by the CCDI. The judges officially pronounced the sentencing and punishment, which was typically a combination of removal from all official posts, prohibition from serving in any future official posts, confiscation of ill-gotten assets, fines, and jail time. The officials charged with corruption in our sample were given as little as dismissal and as much as life imprisonment.

⁴ See <http://cpc.people.com.cn/GB/64093/64387/4671315.html> (accessed March 20, 2019).

⁵ The Organization Department is equivalent to the human resources department of the Chinese Communist Party. Our specific search procedure is as follows. We search for the key words "technology bureau," and "technology head" from the mentioned sources and general web portals. From these records, we construct a database of the names of the technology bureau heads for each province during our sample period. We then identify departure dates.

⁶ Firms report the regional distribution of their revenues in the annual reports. Chinese provinces are grouped into eight geographic regions in China: North (华北), South (华南), Middle (华中), East (华东), North-East (东北), North-West (西北), South-West (西南), and Hong Kong/Macau/Taiwan (港, 澳, 台). Other countries are coded as one separate region.

⁷ Our results are robust to including additional control variables, such as firm leverage and operating performance (ROA). They are also robust to using panel regressions rather than industry-year subsample regressions to estimate the abnormal ETC.

⁸ The Chinese headings for these categories are as follows: (1) 科技三项费用; (2) 科技型中小企业创新基金; (3) 中小企业创新资金项目; (4) 技术改造与工业转型升级经费; (5) 产业化经费、以及设备购买、信息化系统、平台建设等其他经费; (6) 科研项目经费; (7) 专利补贴。

⁹ There were three types of Chinese patents: invention patent, utility model patent, and industrial design patent. The invention patents represented the most significant innovations, and thus are our focus here. Our results are also robust when we use the number of Chinese invention patent applications (regardless of approval status), (ultimately successful) U.S. patent applications, and (ultimately successful) Patent Cooperation Treaty (that is, filed with the World Intellectual Property Organization) patent applications as the numerator in calculating *R&D Efficiency*.

¹⁰ One concern is that due to the lumpiness of patent applications, firms' measured *R&D efficiency* could be artificially volatile. We used firms' foreign sales as a percentage of total sales as an alternative, more continuous measure (albeit very noisy) of *R&D efficiency*, and our main results were robust, albeit slightly weaker than our baseline.

¹¹ We use U.S. patent and citation data to measure firms' future innovation because we anticipate that U.S. patents better capture true innovation than Chinese patents. This deviates from our choice of Chinese patent data in the measurement of firms' *R&D efficiency*. We make this choice because *R&D efficiency* is a key input for subsidy decisions. We reason that Chinese officials making these decisions were more likely to focus on Chinese patents. There are three types of U.S. patents: utility patents, design patents, and plant patents. Our measures are based on the utility patents, which are the most innovative type of patents in the U.S. system. Our results are qualitatively unchanged if we use firms' Chinese invention patents counts or citations to construct these measures instead.

¹² Among firms that patent in the United States, 79.3% (78.8%) of firms were affected by the removals of top provincial officials on corruption charges (the departures of provincial technology bureau heads.) Among firms that patent in China, 79.5% (77.5%) of firms were affected by the removals of top provincial officials on corruption charges (the departures of provincial technology bureau heads.)

¹³ We note an interesting parallel with finding in the United States that there is relatively little campaign giving and high returns to lobbying (Ansolabehere et al. 2003).

¹⁴ U.S. National Science Foundation, Business Research and Development and Innovation: 2013 Detailed Statistical Tables, report NSF 16-313, <https://www.nsf.gov/statistics/2016/nsf16313/pdf/nsf16313.pdf> (accessed April 15, 2019), table 19.

¹⁵ Tabulated results can be found in the online appendix, Tables IA3 and IA4.

¹⁶ The standard deviation of *R&D Efficiency* is 0.117 (Table 3). The coefficient of *R&D Efficiency* is 0.004 in model (1). Therefore, the associated impact on *Subsidies/Sales* is 0.0005 (0.117×0.004), which is 9.4% of the average *Subsidies/Sales* (0.005; Table 3).

¹⁷ The calculation parallels that stated previously: $0.007 \times 0.058 / 0.005 = 8.1\%$.

¹⁸ In an unreported robustness analysis, we constructed binary versions of firms' merit and corruption. *High Efficiency* was equal to one if a firm's *R&D Efficiency* was above the sample median in a given year and zero otherwise. *High AETC* was equal to one if a firm's *AETC/Sales* ratio was above the sample median in a given year and zero otherwise. Our main results were qualitatively similar when using these binary specifications of the key variables. Our results were also robust to de-meaning the control variables by subtracting the industry-year mean for each control variable.

¹⁹ The 56% is calculated as follows: The coefficient on the interaction term between *Subsidies/Sales* and the *Post Removal* indicator is 0.007 in column (4). The standard deviation of *Subsidies/Sales* is 0.008 (Table 3). Thus, the magnitude of the associated increase in *Patents/Sales* is 0.008×0.007 . Because the average value of *Patents/Sales* is 0.0001 (Table 3), in percentage terms the increase is $0.008 \times 0.007 / 0.0001 = 56\%$.

²⁰ We tabulate our *RZ* measure for Chinese industries in the online appendix, Table IA7; in the same table, we list the corresponding

RZ measure for U.S. industries as a reference. The two classifications largely agree, with just a few divergences.

²¹ Bank asset information is obtained from China Stock Market and Accounting Research database (CSMAR) and information on venture presence is obtained from WIND. Ideally, we would have liked to use VC firms' capital under management, but this information was frequently missing. We therefore use the number of VC firms instead.

²² We restricted the sample to just the three years around the cutoff year of 2012 to reduce the overlap with the corruption events used in our baseline analysis.

²³ We used the median corruption index calculated as the number of government officials prosecuted for corruption in a province each year scaled by the province's population (measured in millions of people) over the three-year period prior to the anticorruption campaign to divide Chinese provinces into high- and low-corruption samples. Table IA8 in the online appendix tabulates provincial corruption index scores.

²⁴ This sample is different from our sample of removals of top provincial officials on corruption charges (which includes vice governors and vice party secretaries). Here our focus is on non-corruption-related rotations, and we deem that the relationships with governors and party secretaries to be more valuable for firms than with vice officials.

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